TATE RESOLUTIONS WITH FINITE GENERATORS PER DEGREE UNDER PSEUDO-COHERENCE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; the reuse is declared 'in exactly the same way as Lemma lemma-tate-resolution' with the new twist displayed in full.

Allowed standard prerequisites (editorial).

- Tate resolution construction. lemma-tate-resolution (same chapter, named). Supplies the inductive skeleton, which this proof reuses verbatim.
- Pseudo-coherent complexes. more-algebra-lemma-complex-pseudo-coherent-modules (cross-chapter, named). Certifies the displayed complex is pseudo-coherent.
- Finite cohomology of pseudo-coherent complexes. more-algebra-lemma-finite-cohomology (cross-chapter, named). Yields that H_m is a finite R[x]-module, the one new ingredient.

Source reference. Original lemma 09PN: sources/stacks-dpa/pinned-original-source.tex, source lines 1424–1424. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 067H: focused-seeds/geometry/S9-dpa-tate-resoluton-pseudo-coherent-ring-map/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 064Y: focused-seeds/geometry/S9-dpa-tate-resoluton-pseudo-coherent-ring-map/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 064S: focused-seeds/geometry/S9-dpa-tate-resoluton-pseudo-coherent-ring-map/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $R \to S$ be a pseudo-coherent ring map (More on Algebra, Definition 067H). Then Lemma 09PN holds, with the resolution A of S having finitely many generators in each degree.*

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. This is proved in exactly the same way as Lemma 09PN. The only additional twist is that, given $A(m) \to S$ we have to show that $H_m = H_m(A(m))$ is a finite $R[x_1, \ldots, x_m]$-module (so that in the next step we need only add finitely many variables). Consider the complex

$$\ldots \to A(m)_{m-1} \to A(m)_m \to A(m)_{m-1} \to \ldots \to A(m)_0 \to S \to 0$$

Since S is a pseudo-coherent $R[x_1, \ldots, x_n]$-module and since $A(m)_i$ is a finite free $R[x_1, \ldots, x_n]$-module we conclude that this is a pseudo-coherent complex, see More on Algebra, Lemma 064Y. Since the complex is exact in (homological) degrees $> m$ we conclude that H_m is a finite R-module by More on Algebra, Lemma 064S. $\square$